

Topology First-Year Exam, August 2025, Part 2

For the first five problems, show all your work and support all statements.

1. Suppose that the continuous maps $f, g : X \rightarrow Y$ are homotopic and that the continuous maps $h, k : Y \rightarrow Z$ are homotopic. Show that $h \circ f$ is homotopic to $k \circ g$.
2. Show that the fundamental group of the circle is isomorphic to the additive group of integers. That is, $\pi_1(S^1, 1) \cong \mathbb{Z}$.
3. State and prove the Brouwer fixed point theorem for the unit disk, D^2 .
4. This problem has two parts.
 - (a) Define the Hawaiian earring space, H .
 - (b) Show the images of the generators of the fundamental groups of the circles do not generate the fundamental group of H .
5. This problem has two parts.
 - (a) Define surface.
 - (b) Compute the fundamental group of the connected sum of a torus and a second torus $T^2 \# T^2$.

Answer the following with complete definitions or statements or short proofs.

6. State the Urysohn Lemma.
7. Let E and B be topological spaces. Define what it means for a function $p : E \rightarrow B$ to be a covering map.
8. If $j : A \hookrightarrow X$ is the inclusion of a retract, then show that the induced map on fundamental groups is injective.
9. Let α be a path in X from x_0 to x_1 . Define the map $\hat{\alpha} : \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$ and show that $\hat{\alpha}$ is a group isomorphism.
10. State the Seifert–van Kampen Theorem.
11. Let S^n be the n -sphere. Prove that $\pi_1(S^n)$ is trivial for $n \geq 2$.
12. State the Borsuk–Ulam Theorem for S^2 .
13. State the Tychonoff theorem.
14. **Warning: Suspected error.** The source clearly displays $\pi(X)$, though $\pi_1(X)$ may have been intended; the source notation has been retained.
Describe a space X with $\pi(X) \cong \mathbb{Z}/3\mathbb{Z}$.
15. Define Stone–Čech compactification.